

Latent Semantic Analysis.

Dual mode Factor Analysis.

SVD
Singular Value Decomposition.

Unimode

$$\begin{matrix} f_1 & f_2 & \dots & f_n \\ f_1 & \begin{bmatrix} & & & \end{bmatrix} & \begin{bmatrix} & & & \end{bmatrix} \\ f_2 & & & \\ \vdots & & & \\ f_n & & & \end{matrix}$$

Covariance matrix

$$\begin{matrix} t_1 & \dots & t_n \\ t_1 & \begin{bmatrix} & & & \end{bmatrix} & \begin{bmatrix} & & & \end{bmatrix} \\ t_2 & & & \\ \vdots & & & \\ t_n & & & \end{matrix}$$

$$\text{cat} : \begin{bmatrix} 1 & 1 & 0 & \dots & 1 & \dots & 1 \\ 1 & & & & 1 & & 1 \end{bmatrix}$$

$$\begin{matrix} \text{cat} & \text{dog} & \text{chair} & \text{query} \\ \text{cat} & \begin{bmatrix} 1 & 0.8 & 0.1 \\ & 1 & \\ & & 1 \end{bmatrix} & \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \\ \text{dog} & & & \\ \text{chair} & & & \end{matrix}$$

Query Expansion.

$$\vec{q'} = M \vec{q}$$

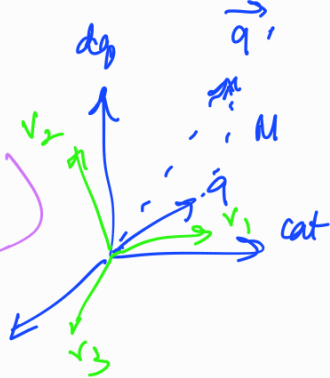
$$= M (\alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 + \alpha_3 \vec{v}_3)$$

M

$$M \rightarrow \begin{cases} \vec{v}_1 \\ \vec{v}_2 \\ \vec{v}_3 \end{cases}$$

$$\begin{matrix} \text{Eigenvectors.} \\ M \vec{v}_1 = \lambda_1 \vec{v}_1 \\ M \vec{v}_2 = \lambda_2 \vec{v}_2 \\ M \vec{v}_3 = \lambda_3 \vec{v}_3 \end{matrix}$$

char



$$= \alpha_1 \lambda_1 \vec{v}_1 + \alpha_2 \lambda_2 \vec{v}_2 + \alpha_3 \lambda_3 \vec{v}_3$$

$$\lambda_1 \gg \lambda_3$$